INSURANCE ANALYSIS DASHBOARD REPORT

---

1. ABSTRACT

This project focuses on analyzing insurance data using Power BI to generate meaningful insights about policies, claims, and customer behavior. The dashboard provides a comprehensive overview of insurance performance across different categories such as policy types, regions, and claim status.

The main objective of this project is to understand claim patterns, identify high-risk areas, and analyze policy distribution. Key metrics such as total policies, total claims, claim amount, and approval rates are highlighted for better decision-making.

This project demonstrates how data visualization can transform raw insurance data into actionable insights for business growth and risk management.

---

2. INTRODUCTION

The insurance industry deals with large volumes of data related to customers, policies, and claims. Analyzing this data is essential for improving services and minimizing risks.

This project aims to develop an interactive Power BI dashboard that provides insights into insurance operations. It helps in understanding claim trends, policy performance, and customer segmentation.

Power BI allows users to explore data dynamically using filters and visualizations, making analysis efficient and user-friendly.

---

3. OBJECTIVES

- To analyze insurance policy and claim data
- To identify claim patterns and trends
- To compare performance across regions
- To evaluate claim approval rates
- To create an interactive dashboard
- To support data-driven decision-making

---

4. DATA ANALYSIS & PROCESSING

The dataset includes information such as policy ID, customer details, premium amount, claim amount, claim status, and region.

Data processing steps include:

- Cleaning and organizing data
- Handling missing values
- Categorizing claim status (approved, rejected, pending)
- Calculating key metrics like total claims and premium

These steps ensure the dataset is accurate and ready for visualization.

---

5. DASHBOARD OVERVIEW

5.1 Key Performance Indicators (KPIs)

- Total Policies
- Total Claims
- Total Premium Amount
- Total Claim Amount
- Claim Approval Rate

5.2 Visualizations Used

- **Bar Chart**: Policies by type
- **Pie/Donut Chart**: Claim status distribution
- **Line Chart**: Claims trend over time
- **Map Visualization**: Region-wise policy distribution
- **Stacked Bar Chart**: Approved vs rejected claims
- **Table**: Customer and policy details
- **Slicers**: Policy type, region, and claim status filters

---

6. EXPLORATORY DATA ANALYSIS

6.1 Policy Distribution

Different policy types contribute differently to total revenue.

6.2 Claim Analysis

Claims are categorized into approved, rejected, and pending.

6.3 Regional Analysis

Certain regions have higher claim rates.

6.4 Time-based Trends

Claims vary over time, indicating seasonal or risk patterns.

---

7. KEY INSIGHTS

- Some policy types generate more claims
- Claim approval rates vary across categories
- High-risk regions show more claim activity
- Premium and claim amounts are closely related
- Trends help identify potential risks

---

8. ADVANTAGES

- Easy visualization of insurance data
- Interactive dashboard with filters
- Helps in risk analysis
- Supports business decision-making
- Scalable for large datasets

---

9. LIMITATIONS

- Limited real-time data
- No predictive analytics
- Data accuracy depends on source
- Limited customer-level insights

---

10. FUTURE ENHANCEMENTS

- Add real-time data integration
- Implement predictive analytics
- Improve dashboard interactivity
- Add fraud detection analysis
- Deploy on web platform

---

11. CONCLUSION

The Insurance Analysis Dashboard successfully demonstrates how Power BI can be used to analyze insurance data effectively.

It provides valuable insights into policy performance, claim patterns, and risk areas, helping organizations make informed decisions. With further enhancements, it can become a powerful tool for insurance analytics.